

Arrow of Time and Entropy

PAX L5 Narrow L2 General 27B | Anticloud FZ LLE

Subject: Proposed Resolution to the Arrow of Time via Cosmological Initial Conditions and Gravitational Phase Space Suppression

Model: PAX L5 Narrow L2 General 27B (Anticloud FZ LLE)

Status: Rigorous Theoretical Proposal

1. Core Hypothesis: The Low-Entropy Singularity as a Gravitational Vacuum State

The central paradox of the Arrow of Time is that while microscopic physical laws (Newtonian, Quantum, Relativistic) are predominantly time-symmetric (T -invariant), macroscopic thermodynamics exhibits a strict irreversibility defined by the Second Law ($dS/dt \geq 0$). The standard resolution posits that the universe began in an exceptionally low-entropy state (the Past Hypothesis). However, this has historically been treated as a brute fact without a dynamical mechanism explaining *why* such a specific microstate was selected.

Hypothesis: The Big Bang did not represent a random selection from the phase space of possible initial conditions but rather the unique ground state of a **Gravitational Vacuum**. In this framework, "low entropy" corresponds to a state where gravitational degrees of freedom are maximally suppressed due to extreme spacetime curvature constraints, effectively freezing the system into a single macro-state with vanishing micro-state degeneracy. The arrow of time emerges dynamically as gravity drives the expansion of the accessible phase space volume (Ω) through gravitational clumping and horizon formation, transitioning the universe from a state of constrained vacuum fluctuations to a high-entropy thermal equilibrium dominated by black hole radiation and dark energy dilution. Time asymmetry is not a boundary condition imposed on the equations of motion but an emergent property of the **phase space topology** evolving under the influence of non-linear general relativity.

2. Mathematical Framework

We construct a formalism combining Penrose's Weyl Curvature Hypothesis with modern holographic entropy bounds and Liouville dynamics in the context of the Friedmann-Lemaître-Robertson-Walker (FLRW) metric perturbed by quantum fluctuations.

A. The Entropy Functional

Let $\mathcal{M}$ be the spacetime manifold with metric $g_{\mu\nu}$. We define the total entropy S_{tot} as the sum of matter entropy S_m and gravitational entropy S_g :

$$S_{\text{tot}}(t) = S_m(t) + S_g(t)$$

Following Penrose, we identify gravitational entropy with the integral of the squared Weyl tensor

$C_{\alpha\beta\gamma\delta}$ over a spatial hypersurface Σ_t :

$$S_g[\Sigma_t] \propto \int_{\Sigma_t} d^3x \sqrt{h} \left(C_{\alpha\beta\gamma\delta} C^{\alpha\beta\gamma\delta} \right)^n$$

Where h is the induced metric determinant. For the early universe (FLRW limit), the Weyl tensor vanishes ($C=0$), implying $S_g \approx 0$, satisfying the low-entropy requirement.

B. Phase Space Volume Dynamics

Let $\Gamma(t)$ be the available phase space volume at cosmic time t . The Boltzmann entropy is $S = k_B \ln |\Gamma|$.

We propose a modified Liouville theorem for expanding spacetimes coupled to gravity. The evolution of the phase space density $\rho(x,p,t)$ obeys:

$$\frac{D\rho}{Dt} + \rho \nabla \cdot v_{\text{eff}} = \sigma_{\text{gen}}$$

Where σ_{gen} represents the generation of coarse-grained entropy due to the breakdown of integrability in non-linear gravitational interactions (chaos).

In the early universe, the Hamiltonian flow is constrained by the Planck-scale cutoff and the vanishing Weyl tensor, forcing the system onto a lower-dimensional submanifold $\mathcal{M}_0 \subset \Gamma_{\text{total}}$. As the universe expands, the effective potential landscape flattens, allowing trajectories to explore higher dimensions of Γ .

C. The Measure Problem Resolution

To avoid the measure problem (where high-entropy states vastly outnumber low-entropy ones), we introduce a weight function $W(\psi)$ based on the action principle for the initial wavefunction of the universe $|\Psi_U\rangle$:

$$\langle \Psi_U | e^{-iHt} | \Psi_U \rangle \implies P(\text{State}) \propto e^{-I[\text{State}]}$$

Where I is the Euclidean action. We posit that the path integral is dominated by configurations where the conformal factor ϕ satisfies $\Box \phi = 0$ at the singularity, uniquely selecting the isotropic, homogeneous vacuum.

3. Proof Sketch / Key Steps

****Step 1: Establishing the Vanishing Weyl Limit****

Demonstrate that in the limit $a(t) \rightarrow 0$ (scale factor approaches zero), the Einstein Field Equations $G_{\mu\nu} = 8\pi G T_{\mu\nu}$ force the Weyl tensor to vanish faster than any matter perturbations can grow, provided the initial data is smooth and isotropic. This mathematically enforces $S_g(t_0) \rightarrow 0$.

Key Logic: If $R_{\mu\nu} \sim \rho g_{\mu\nu}$ dominates near the singularity, then $C_{\mu\nu\rho\sigma} \rightarrow 0$ is a fixed point of the evolution equations.

****Step**